$\boxed{a(\text{circular disk})=\pi r^2}$

1. Area of a circular disk

$S=\{(x,y)\mid x^2+y^2\le r^2\}\Rightarrow A(r):=a(S)\ \exists$, $A(r)=r^2A(1)=\pi r^2$

proof: ① let $g(x)=\sqrt{r^2-x^2}$, $f(x)=-g(x)$ defined on $[-r,r]$.

② ∵ f, g bounded and piecewise monotonic ∴ $I(f), I(g)\ \exists$.

∴ $a(S)\ \exists$, $a(S)=I(g-f)$

③ $A(r):=a(S)=\int_{-r}^{r}(g-f)\,dx$

$=2\int_{-r}^{r}\sqrt{r^2-x^2}\,dx$

∴ $A(1)=2\int_{-1}^{1}\sqrt{1-x^2}\,dx$

$\pi:=A(1)$

④ $A(r)=2\int_{-r}^{r}g(x)\,dx=2\frac{1}{k}\int_{-kr}^{kr}g(\frac{x}{k})\,dx$

$=2r\int_{-1}^{1}g(rx)\,dx$, $k=\frac{1}{r}$

$=2r\int_{-1}^{1}\sqrt{r^2-(rx)^2}\,dx$

$=2r^2\int_{-1}^{1}\sqrt{1-x^2}\,dx$

$=r^2A(1)=\pi r^2$

2. $S=\{(ax,by)\mid x^2+y^2\le 1\}$

$S'=\{(x',y')\mid (\frac{x'}{a})^2+(\frac{y'}{b})^2\le 1\}$, $a,b>0$

$\Rightarrow S=S'$

proof: ∵ $\exists T, T^{-1}$, $(x',y')=T(x,y)=(ax,by)$,

$(x,y)=T^{-1}(x',y')=(\frac{x'}{a},\frac{y'}{b})$

又 ∵ $T_1(x,y)=(ax,by)=(x',y')=T_2(x',y')$

here $X_1=\{(x,y)\mid x^2+y^2\le 1\}$

$X_2=\{(x',y')\mid (\frac{x'}{a})^2+(\frac{y'}{b})^2\le 1\}$

∴ $S=T_1X_1=T_2X_2=S'$ (by 02184.0.2)

3. Area of a elliptical disk

$S=\{(ax,by)\mid x^2+y^2\le 1\}$

$\Rightarrow A(a,b):=a(S)\ \exists$, $A(a,b)=\pi ab$

proof: ① ∵ $S=\{(x,y)\mid (\frac{x}{a})^2+(\frac{y}{b})^2\le 1\}$ (from 2)

let $g(x)=b\sqrt{1-(\frac{x}{a})^2}$, $f(x)=-g(x)$ defined on $[-a,a]$.

∴ $S=\{(x,y)\mid -a\le x\le a, f(x)\le y\le g(x)\}$

② ∵ f, g bounded, piecewise monotonic.

∴ $I(f), I(g)\ \exists$, 又 ∵ $f\le g$ ∴ $a(S)\ \exists$, $a(S)=I(g-f)$

③ $A(a,b):=a(S)=2I(g)=2b\int_{-a}^{a}\sqrt{1-(\frac{x}{a})^2}\,dx$

$=\frac{2b}{k}\int_{-ka}^{ka}\sqrt{1-(\frac{x}{ka})^2}\,dx=2ab\int_{-1}^{1}\sqrt{1-x^2}\,dx$, $(k=\frac{1}{a})$

∴ $A(a,b)=\pi ab$

4. Area of an ordinate set under a similarity transformation

$f\ge 0$, integrable on $[a,b]$, S its ordinate set, $k>0$.

$\Rightarrow a(kS)\ \exists$, $a(kS)=k^2a(S)$

proof: ① ∵ S ordinate set of f on $[a,b]$

∴ kS ordinate set of a fun, say g, on $[ka,kb]$.

② ∵ $y=g(x)\Leftrightarrow \frac{y}{k}=f(\frac{x}{k})$

∴ $g(x)=y=kf(\frac{x}{k})$

③ ∵ $I(f,[a,b])\ \exists$ ∴ $I(f(\frac{x}{k}),[ka,kb])\ \exists$

∴ $I(g)\ \exists$ ∴ $a(kS)\ \exists$.

∴ $a(kS)=I(g)=\int_{ka}^{kb}g(x)\,dx=k\int_{ka}^{kb}f(\frac{x}{k})\,dx$

$=k^2\int_a^b f(x)\,dx=k^2a(S)$.

5. $f\ge 0$, integrable on $[a,b]$,

S its ordinate set, $S'=\{(k_1x,k_2y)\mid (x,y)\in S\}$, $k_1,k_2>0$

$\Rightarrow a(S')\ \exists$, $a(S')=k_1k_2a(S)$.

proof: ① ∵ S ordinate set of f on $[a,b]$

∴ S' ordinate set of a fun, say g, on $[k_1a,k_1b]$

② ∵ $y=g(x)\Leftrightarrow \frac{y}{k_2}=f(\frac{x}{k_1})$ ∴ $g(x)=k_2f(\frac{x}{k_1})$

③ $I(f,[a,b])\ \exists\Rightarrow I(f(\frac{x}{k_1}),[k_1a,k_1b])\ \exists$, $\Rightarrow I(g)\ \exists$.

∴ $a(S')\ \exists$, $a(S')=I(g)=\int_{k_1a}^{k_1b}g(x)\,dx$

∴ $a(S')=k_2\int_{k_1a}^{k_1b}f(\frac{x}{k_1})\,dx=k_1k_2\int_a^b f(x)\,dx=k_1k_2a(S)$

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